Spatial relationship between oil infrastructure and conflict in the Middle East

Erika Yazmín Blanco
Felipe Manzi
Olta Reçica
Olsa Berani

2026-03-27

Contents

1 Introduction

The spatial distribution of armed conflict is shaped by a combination of political, economic, and geographic factors. In regions such as the Middle East, where oil infrastructure plays a central strategic role, understanding how conflict events are distributed relative to these assets is particularly relevant.

Oil infrastructure, including extraction sites, pipelines, and refineries, represents key economic and geopolitical resources. While these assets are often discussed in relation to conflict dynamics, the extent to which conflict events are spatially concentrated around them remains an empirical question.

This project examines whether systematic spatial patterns exist between oil infrastructure and conflict events across eight countries in the Middle East: Saudi Arabia, Iraq, Iran, Kuwait, United Arab Emirates, Qatar, Syria, and Yemen. Specifically, the research question guiding this analysis is: **What is the spatial relationship between oil infrastructure and armed conflict events in the Middle East?**

The analysis evaluates whether proximity-based relationships can be observed using multiple geospatial approaches. Specifically, the study combines city-level distance measures, buffer-level groupings, and conflict-level proximity metrics to provide a comprehensive spatial assessment. The analysis focuses on fixed oil infrastructure, such as plants and processing facilities, treated as discrete strategic assets. Linear infrastructure, such as pipelines, is excluded due to its extensive spatial coverage, which would reduce the analytical precision of proximity-based measures.

2 Motivation and literature Review

Natural resource infrastructure has long been associated with geopolitical tension and strategic competition. In the Middle East, oil infrastructure is deeply embedded within broader political and economic systems, often overlapping with areas of territorial contestation and military activity.

Existing research suggests that strategic assets may influence the geography of conflict, either directly through targeting or indirectly through regional instability. However, these relationships are complex and often operate at broader spatial scales rather than at precise locations.

A key limitation in the literature is the reliance on single measures of proximity, which may fail to capture the multi-dimensional nature of spatial relationships. Conflict may not occur exactly at infrastructure sites but may still be systematically closer to them compared to other areas.

To address this gap, this project adopts a multi-method geospatial approach, allowing for the identification of consistent spatial patterns across different measurement strategies.

3 Data and Methodology

3.1 Data

The analysis focuses on eight countries in the Middle East: Saudi Arabia, Iraq, Iran, Kuwait, United Arab Emirates, Qatar, Syria, and Yemen. To examine the spatial relationship between oil infrastructure and conflict, the project combines georeferenced data on conflict events, fixed oil and gas infrastructure, populated places, and country boundaries.

Conflict data come from the Armed Conflict Location & Event Data Project (ACLED), using an aggregated weekly dataset for the Middle East. From the raw file, the sample is restricted to the eight study countries. Observations with missing geographic coordinates are removed, and the date variable is standardized. The remaining records are converted into a spatial dataset using the reported longitude and latitude coordinates. Because the source is aggregated at the weekly level, each observation represents a conflict record associated with a specific location and week.

Oil infrastructure is measured using the Global Oil and Gas Plant Tracker (GOGPT). This dataset provides point locations for oil and gas plants and units, together with information such as plant name, operational status, fuel classification, ownership, and capacity. The data are filtered to the study countries, and only observations with valid latitude and longitude are retained. These plant locations constitute the main infrastructure layer used in the empirical analysis.

To represent populated locations, the study uses Natural Earth populated places data. These data provide city point locations and are filtered to the same eight countries. They are used in the city-level analysis, where each city is linked to the nearest conflict event and classified according to its proximity to oil infrastructure. Country boundaries are taken from Natural Earth administrative polygons. These boundaries are used to define the study area and to provide geographic context for the maps.

All spatial datasets are stored in two coordinate reference systems. Data in EPSG:4326 are used for mapping and visualization, while data in EPSG:3857 are used for buffer construction and distance calculations. This ensures consistency between cartographic presentation and spatial measurement.

3.2 Methodology

The analysis combines three complementary approaches to evaluate the spatial relationship between oil infrastructure and conflict: buffer-based measures distance-based measures group comparisons

This triangulation strengthens the robustness of the findings by identifying patterns that are consistent across different methodological approaches.

3.2.1 Buffer Analysis and Spatial Classification

First, we constructed buffer zones of 30 km and 60 km around oil infrastructure sites.

Cities are classified based on whether their geographic coordinates fall within these buffer areas. A city is considered to be "within a buffer" if its location intersects with the area defined by the buffer distance around any oil infrastructure site. In practical terms, this means that the city lies within a specified radius (30 km or 60 km) of at least one oil-related location.

Based on this criterion, cities are grouped into two categories:

cities near oil infrastructure (located within the buffer zones) cities farther from oil infrastructure (located outside the buffer zones)

This classification allows for a comparison between cities that are geographically close to oil infrastructure and those that are not, providing a basis for analyzing differences in their proximity to conflict events.

Similarly, conflict events are classified according to whether they occur within these buffer zones. This allows for the calculation of the share of conflict events located near oil infrastructure.

3.2.2 City-Level Distance to Conflict

For each city, the distance to the nearest conflict event is calculated using spatial nearest-neighbor methods. Specifically, each city is matched to its closest conflict event, and the geographic distance between the two is computed. This produces a continuous measure of proximity to conflict for every city, allowing for a detailed comparison across locations rather than relying on simple binary classifications.

Cities are then grouped based on their proximity to oil infrastructure, as defined by the buffer zones constructed earlier (e.g., within 30 km or 60 km of oil sites). Within each group, the average distance to the nearest conflict event is calculated. By comparing these averages, the analysis assesses whether cities located near oil infrastructure tend to be systematically closer to conflict events than those located farther away. This approach provides a straightforward way to identify spatial differences in conflict exposure associated with proximity to oil infrastructure.

3.2.3 Conflict-Level Distance to Oil Infrastructure

Distances from each conflict event to the nearest oil infrastructure site are computed using spatial nearest-neighbor techniques. For each event, the minimum distance to any oil-related site is identified, producing a continuous measure of proximity.

This approach generates a full distribution of distances, which can be summarized using descriptive statistics such as the mean, median, and range. Examining this distribution allows for an assessment of how conflict events are spatially positioned relative to oil infrastructure, including whether they tend to cluster within specific distance bands or are more widely dispersed across the region.

4 Results

4.1 Cities Proximity to Conflict

At the 30 km threshold, cities located near oil infrastructure exhibit lower average distances to conflict events compared to those located farther away (approximately 75.6 km vs. 83.6 km). A Welch two-sample t-test confirms that this difference is not statistically significant ($t = 0.73$, $df = 119.3$, $p = 0.469$), suggesting that the modest gap observed at this scale could plausibly arise by chance. At the 60 km threshold, the difference becomes substantially larger. Cities near oil infrastructure are, on average, located approximately 23 km closer to conflict events than those farther away (69.6 km vs. 92.7 km). This difference is statistically significant ($t = 2.26$, $df = 167.6$, $p = 0.025$), indicating that the spatial association between city proximity to oil infrastructure and exposure to conflict is unlikely to be due to random variation alone.

This pattern indicates that the spatial association between oil infrastructure and conflict is more clearly observable when broader geographic definitions of proximity are considered. The 60 km buffer is the threshold at which the signal becomes statistically detectable with this dataset. Importantly, these results reflect differences in spatial distribution rather than direct causal relationships, and should be interpreted as descriptive patterns.

4.2 Conflict Proximity to Oil Infrastructure

The distribution of distances from conflict events to oil infrastructure shows that conflict typically occurs within a regional range of oil-related locations. The median distance is approximately 81 km, with an average close to 88 km, indicating that while some events occur near infrastructure, many are located at moderate distances. This suggests that conflict is not tightly concentrated around oil sites but tends to occur within broader regions where oil infrastructure is present.

Country-level patterns reveal substantial heterogeneity. Kuwait and Qatar have the shortest average distances (25.5 km and 38.2 km respectively), reflecting their small geographic size and high density of oil infrastructure. Iraq, with the largest number of conflict events (22,541), shows an average distance of 52.2 km, indicating meaningful spatial overlap between conflict and oil sites. At the other extreme, Saudi Arabia and Yemen show average distances exceeding 115 km, suggesting that in these countries conflict events are more spatially separated from oil infrastructure.

4.3 Share of Conflict Events Near Oil Infrastructure

Buffer-based results show that:

Approximately 25% of conflict events occur within 30 km of oil infrastructure Approximately 40% of conflict events occur within 60 km

This indicates that a substantial share of conflict events takes place in areas geographically close to oil infrastructure, particularly when larger spatial thresholds are considered. However, the majority of conflict events still occur outside these buffers, reinforcing the idea that conflict is not exclusively concentrated near oil sites.

5 Conclusions and Limitations

This study identifies consistent spatial patterns between oil infrastructure and the distribution of conflict events in the Middle East. Across multiple approaches, results suggest that conflict tends to occur relatively closer to oil infrastructure, particularly when broader distance thresholds are considered. However, these patterns are not localized, and conflict events are distributed across a wide geographic range. However, the observed spatial associations may reflect underlying regional dynamics, such as political instability, geographic clustering, or historical factors, rather than direct relationships between oil infrastructure and conflict occurrence.

Several limitations should be noted:

The analysis does not account for differences in conflict intensity or type Country-level heterogeneity is not explicitly modeled Distance-based measures do not capture strategic or political mechanisms

Future research could extend this approach by incorporating temporal dynamics, conflict intensity measures, or additional covariates to better understand the factors underlying these spatial patterns.

6 References

Add your references here.